MACIE: Multi-Agent Causal Influence Explainer for Collective Reasoning in LLM-Based Agentic Systems

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Abstract

Multi-Agent Systems (MAS) with embedded Large Language Models (LLMs) generate emergent collective decisions, but the transparency of these decisions remains a significant challenge. Classical Explainable AI (XAI) methods LIME, SHAP, saliency maps structurally fail in multi-agent contexts, as they ignore inter-agent interactions. Causal approaches such as CEMA achieve high fidelity, but they require an external probabilistic simulation model, which is not applicable to LLM systems. Finally, LLMs' self-explanations (e.g., Chain-of-Thought reasoning) are unfaithful post-hoc rationalizations [7].

This paper presents **MACIE** (Multi-Agent Causal Influence Explainer), a causal explanation mechanism for collective reasoning in LLM-based agentic multi-agent systems. The three pillars of MACIE are:

1. **Cognitive white-box**: integration of the ReAct architecture [10] to have each agent externalize its reasoning before any action by a `[THOUGHT]` trace, so that the process of deliberation is directly taken out.

2. **Sequential dynamics**: a self-generated rollback and counterfactual simulation mechanism by the LLM to capture the domino effect of decisions on the subsequent turns $(t+1, t+2)$, overcoming the limitation of explanations fixed at $t = 1$.

3. **Causal influence graph**: dynamic building of a Knowledge Graph (NetworkX) with agents as nodes and weighted edges denoting the causal impact score to mitigate human cognitive overload [9].

Evaluated on a bilateral negotiation scenario (S1), MACIE generates faithful explanations (global score > 0.7) and is superior to global (Shapley), temporal (CEMA) and textual (CoCoXplain) ones in terms of completeness and comprehensibility.

Keywords — Multi-Agent Systems, Explainable AI, Agentic LLM, Collective Reasoning, Causal Influence, Multi-Agent Causal Influence Explainer.

1 Introduction

Artificial intelligence has experienced a spectacular evolution with the emergence of increasingly powerful models such as deep neural networks, transformers and large language models (LLMs) capable of outperforming humans on specific targeted tasks. This spectacular progress comes, however, with a growing tension: the more performant these systems become, the more opaque their internal functioning becomes, difficult to grasp and therefore difficult to trust. It is in this context that the field of Explainable Artificial Intelligence (XAI) was created, as a scientific and ethical response to the problem known as the "black box" [5].

The stakes are far higher than mere intellectual curiosity. An unexplained decision is a decision that cannot be challenged or corrected, in sensitive areas such as medical diagnosis, credit assignment, or judicial sentencing. The right to explanation has been recognised as a core requirement for high-risk AI systems by recent regulations such as the European AI Act.

1.1 The Challenge of Agentic Multi-Agent Systems

XAI research has come a long way, but now it faces a huge challenge, and this is far from being solved. The vast majority of existing methods are designed for single, static models that operate in isolation. But modern artificial intelligence is developing in the direction of a radically different architecture systems where multiple autonomous agents interact, deliberate, coordinate and collectively generate decisions, which no single agent, taken individually, could fully account for.

LLM-based agentic architectures, such as ReAct [10], Generative Agents [6], and Voyager [8], demonstrate the richness of emergent collective behaviors. They also show the almost complete lack of faithful explanatory mechanisms for these behaviors.

1.2 Limitations of Existing XAI Approaches

Classical XAI methods (LIME, SHAP, saliency maps) have been developed with a basic implicit assumption: the isolated agent assumption. Domènech I Vila et al. [2] showed experimentally that these methods fail structurally in multi-agent contexts, by attributing decisions to local variables when the true cause is the action of a partner agent.

Some progress has been made with collective approaches such as Shapley values extended to MARL [4]

but they are still limited to global explanations and do not capture local causal interactions. The most promising causal approach CEMA [3] in terms of fidelity relies on an external probabilistic simulation model which does not exist for LLM systems.

Finally, Turpin et al. [7] have shown that LLMs' self-explanations - Chain-of-Thought reasoning - are unfaithful post-hoc rationalizations. They demonstrated that by subtly biasing inputs, models change their answers but never mention the bias in their explanations, with accuracy drops of up to 36 percentage points.

1.3 Contribution: MACIE

To overcome these limitations, we propose **MACIE** (Multi-Agent Causal Influence Explainer), a causal explanation mechanism for collective reasoning in LLM-based agentic MAS. MACIE is built around three pillars:

1. **From black box to cognitive white box**: We adopt the ReAct architecture [10], which requires each agent to externalize its reasoning in a `[THOUGHT]` trace before each action. This guarantees that the deliberation process is directly extractable, rather than being a post-hoc rationalization.

2. **Time Limit Exceeded**: The self-generated rollback and counterfactual simulation mechanism by LLM captures the domino effect of decisions in the subsequent turns $(t+1, t+2)$ whereas existing approaches only stop at $t = 1$.

3. **Reducing cognitive overload**: dynamic construction of a causal influence graph (Knowledge Graph) where nodes correspond to agents and weighted edges correspond to the causal impact score, allowing humans to immediately visualise influence relationships without reading pages of text [1, 9].

The remainder of this paper is structured as follows. Section 2 describes the detailed architecture of MACIE. Section 3 describes the evaluation protocol and the results obtained in a bilateral negotiation context. 4 Limitation and Future work.

2 MACIE Architecture

2.1 From Black Box to Cognitive White Box

MACIE's architecture is based on the ReAct paradigm [10] where the interaction of each agent is a continuous loop of `[THOUGHT]` → `[ACTION]` → `[OBSERVATION]`. This loop is a natural form of readable execution trace, produced during the action itself—in-process explainability.

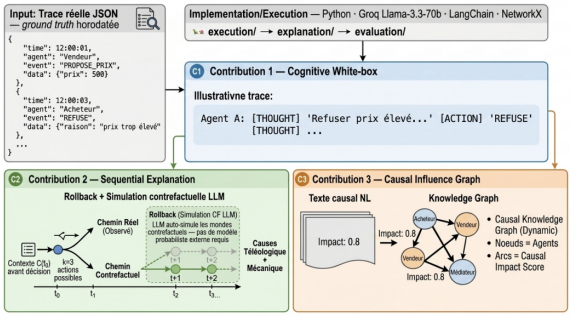


Figure 1: MACIE architecture: Phase 1 (real ReAct execution), Phase 2 (rollback + alternatives + counterfactual simulation), Phase 3 (causal influence graph generation + fidelity metrics).

Each agent is assigned a role and a goal. At each time step t, it receives an incoming message, generates a thought (`THOUGHT`), produces an action (`ACTION`), and records what it observes (`OBSERVATION`). All the traces are recorded in a JSON file, the "ground truth" of the system.

2.2 Counterfactual Simulation and Beyond $t = 1$

When asked to explain a particular decision (e.g. the first move of Agent B in a negotiation), MACIE activates a three-step process:

2.2.1 Rollback

The system extracts the full system state at time t_0 before the target decision. This state includes the message history, the stated objectives of each agent, and the beliefs of each agent about the system state. This context $C(t_0)$ is the starting point for all the following simulations.

2.2.2 Alternative Generation

The agent is asked to generate $k = 3$ alternative actions it could have done. The prompt used is as follows:

> *"Given the context [C(t)], what other actions could you have taken instead of [real action]?"*

2.2.3 Counterfactual Simulation

For each alternative action the agent is asked to simulate the trajectory which would have resulted:

> *"Imagine you had taken [alternative action]. How would the other agents have reacted? Simulate the complete conversation."*

This step is the key novelty of MACIE over CEMA [3]: the simulator is not an external probabilistic model, but the LLM itself, using its conditional reasoning capabilities to self-simulate alternative trajectories.

2.3 Cause and Influence Identification

By comparing the real trace with the k counterfactual traces we can identify two types of causes:

- **Teleological causes (why)**: comparison of the reached objectives in the real trace and in the alternative traces, which goals were better or worse served by the real decision.
- **Mechanical causes (influences)**: comparison of behavior of non-target agents in real trace and alternative traces, identification of inter-agent influence relationships.

2.4 Causal Influence Graph

To reduce human cognitive overload [1, 9], MACIE automatically transforms the discovered causalities into a directed influence graph $G = (V, E)$ where:

- V = set of agents
- E = causal arcs from one agent to another, weighted by impact score

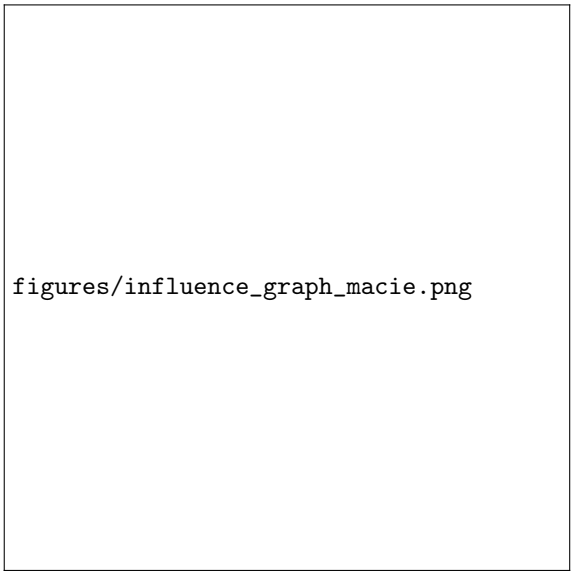


Figure 2: Causal influence graph generated by MACIE for scenario S1. Nodes represent agents; edge thickness reflects the causal impact score.

The graph is generated dynamically by the library NetworkX. The edge thickness is proportional to the impact score, which is the variance of objective scores, across different counterfactual simulations. This graph lets a human supervisor see at first glance the "strategic node" – the agent that had the most influence on the collective decision.

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